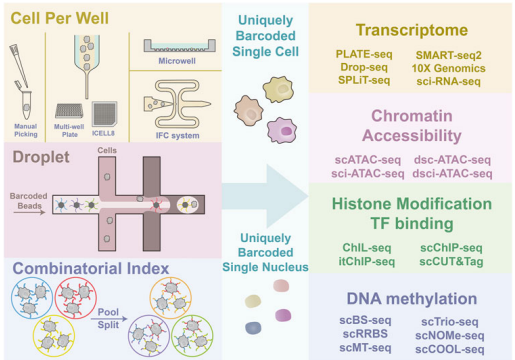
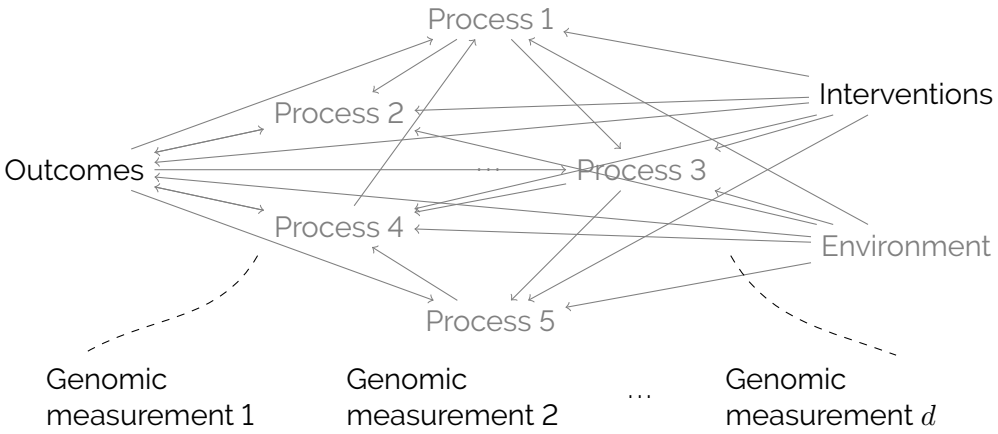
Introduction | Principles

Computational biology tries to understand complex biological systems by analyzing large-scale genomic measurements.



Zhang et al. (2022)

Focus on underlying processes reflected by the genomic measurements.



The key to computational biology is to formulate a good question.



AI will soon be able to do almost everything else.

Which of the following are good computational biology questions?

1. How do reward circuits in the brain work?
2. What cellular processes are activated when a cell becomes stressed?
3. Why are some subtypes of breast cancer more aggressive?
4. Which variables have significantly different distributions between two populations?
5. Given two sets of objects, is the number of objects in common between them higher than would be expected by chance?
6. What is the most accurate prediction rule for a right-censored outcome given a large set of predictors?

Computational biology questions must be both:

- biologically meaningful (“so what?”)
- quantitatively answerable

Descriptions are not automatically meaningful.

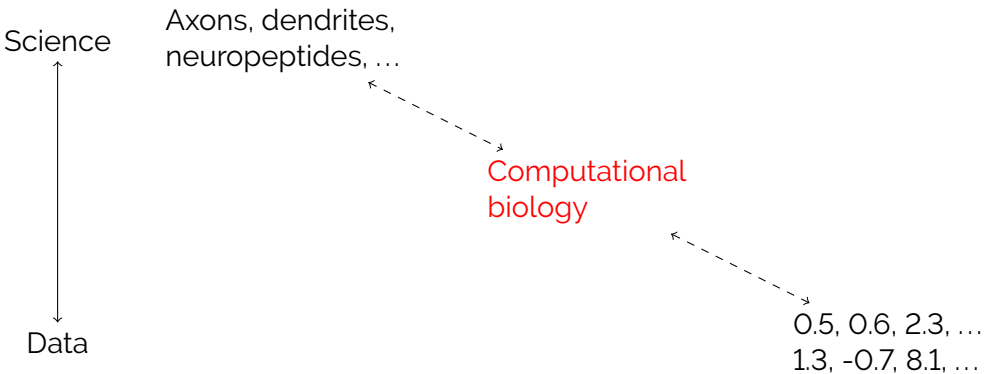


Wikipedia

Descriptions are not automatically meaningful.



Constructing substantively meaningful quantitative constructs is hard because it requires understanding, bridging, and sometimes creating conceptual schemas.



Meaning comes from how genomic measurements covary with other measurements that already carry meaning.

